Diabetes Mellitus

Classification

Diagnosis

Glucose Homeostasis

Pathophysiology

Clinical manifestation

Treatment

Classification

Type 1 diabetes

(ß cell destruction leading to absolute insulin deficiency)

Type 2 diabetes

(Progressive insulin secretion defect
on the background of insulin resistance)

Other specific types of diabetes

(Genetic defect in ß -cell function / insulin action,
Disease of exocrine pancreas,
Drug induced)

Gestational diabetes mellitus

Diagnosis

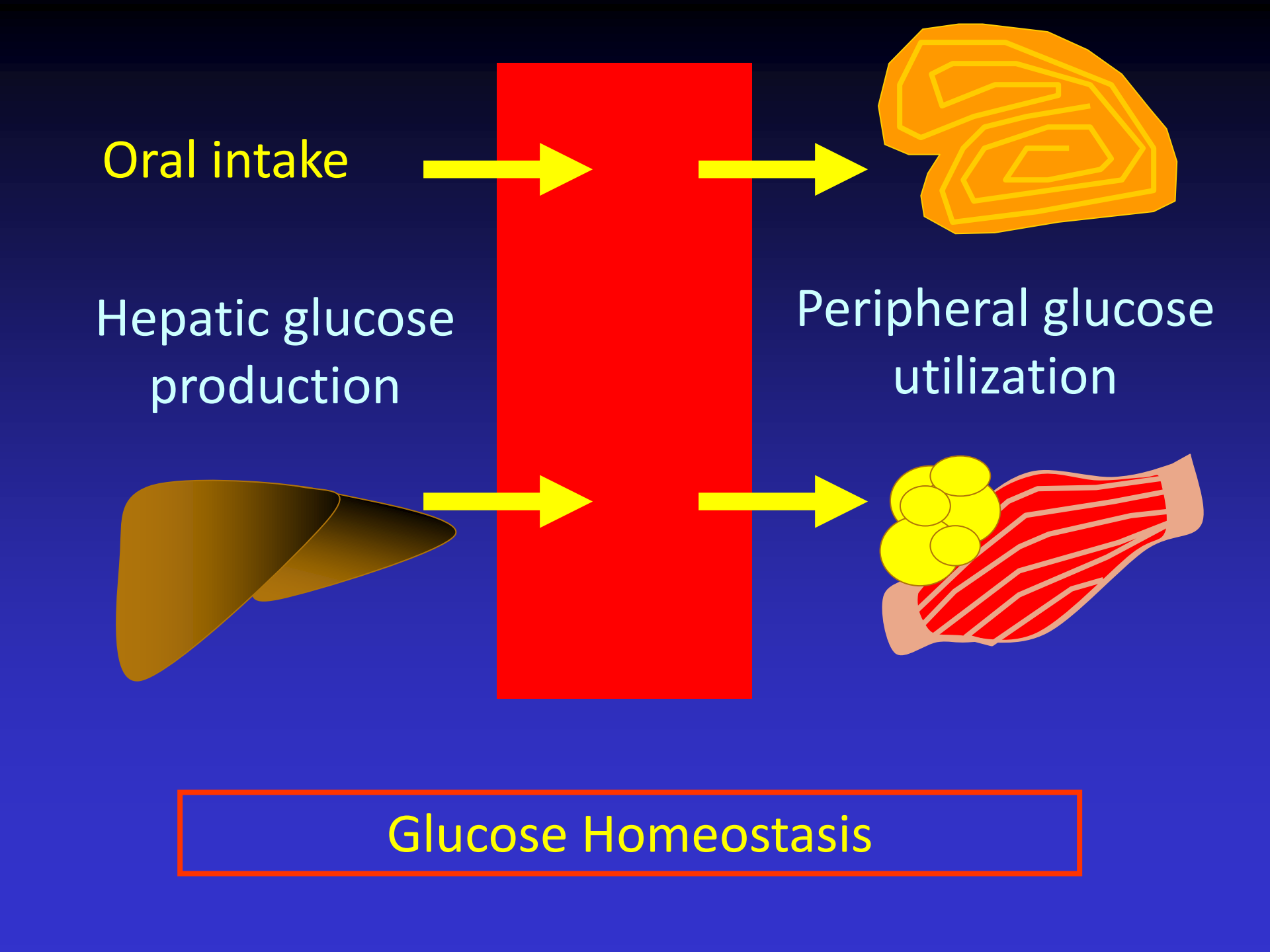
Fasting plasma glucose

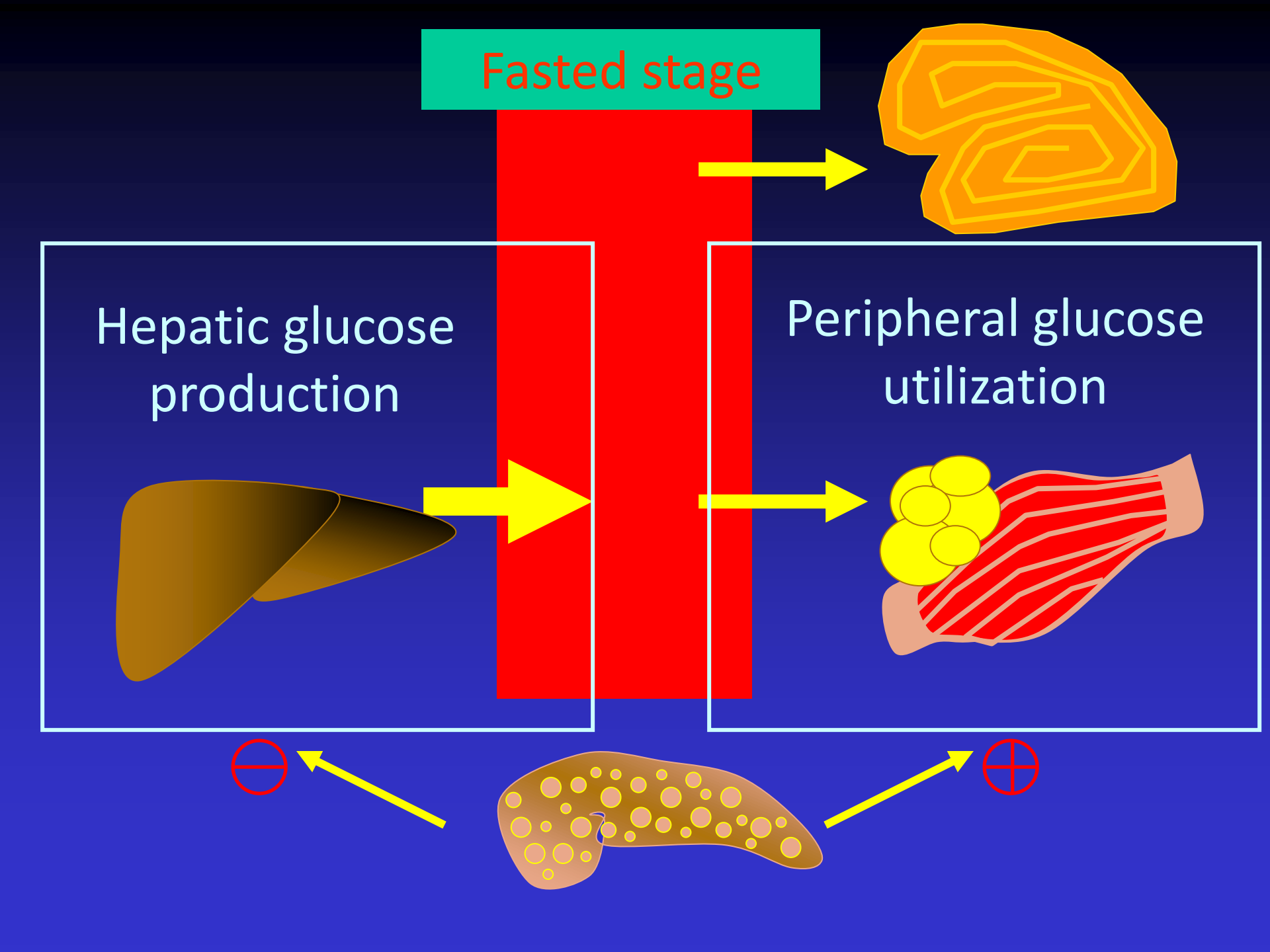
75-g oral glucose tolerance test (75-g OGTT)

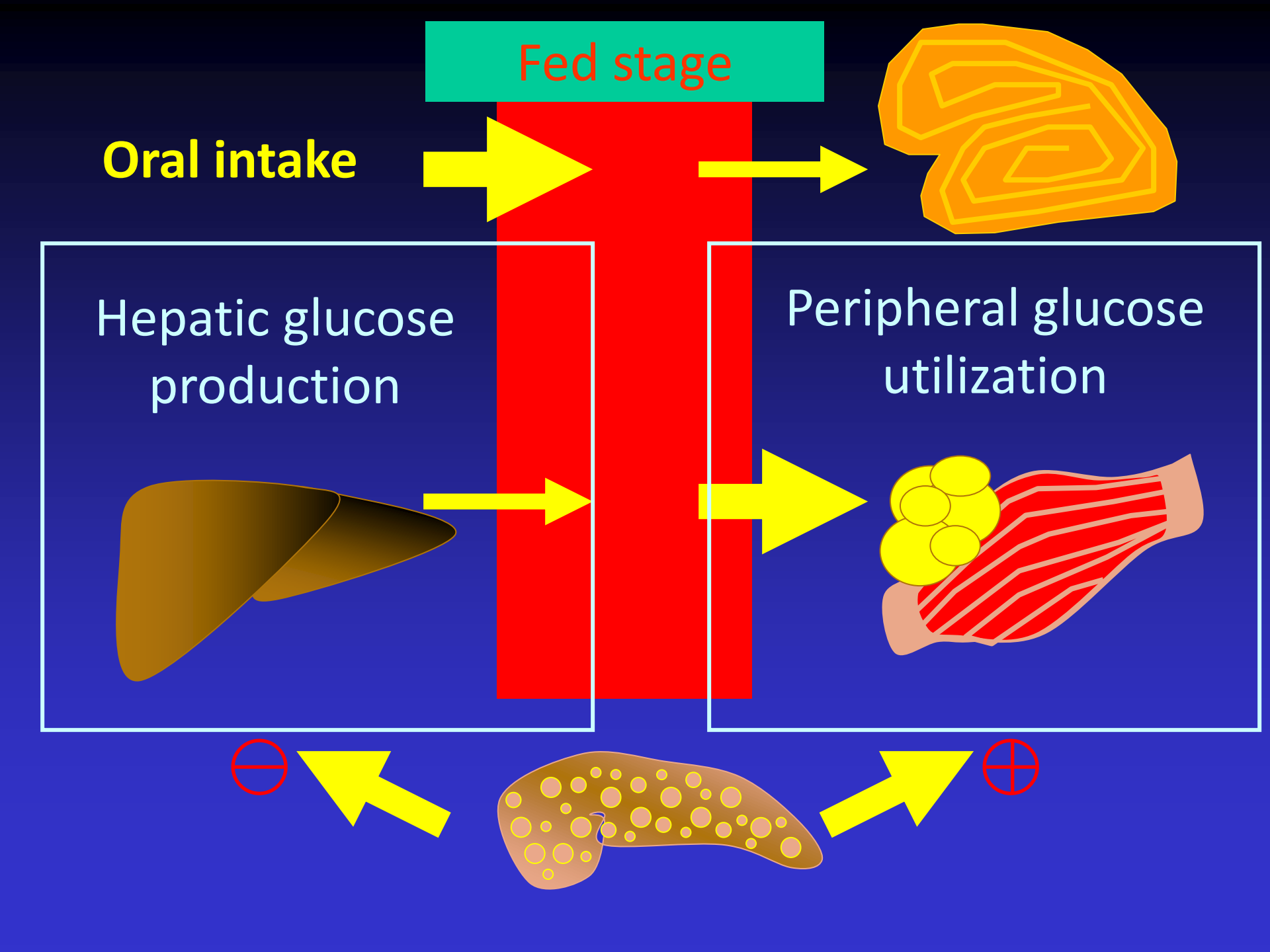
American Diabetes Association

	FPG (mg/dl)	2-h PG (mg/dl)		HbA1c (%)
Normal fasting glucose (NFG)	<100	<140	Normal glucose tolerance (NGT)	<5.7
Impaired fasting glucose (IFG)	100-125	140-199	Impaired glucose tolerance (IGT)	5.7-6.4
Diabetes mellitus (DM)	≥126	≥200	Diabetes mellitus (DM)	≥6.5

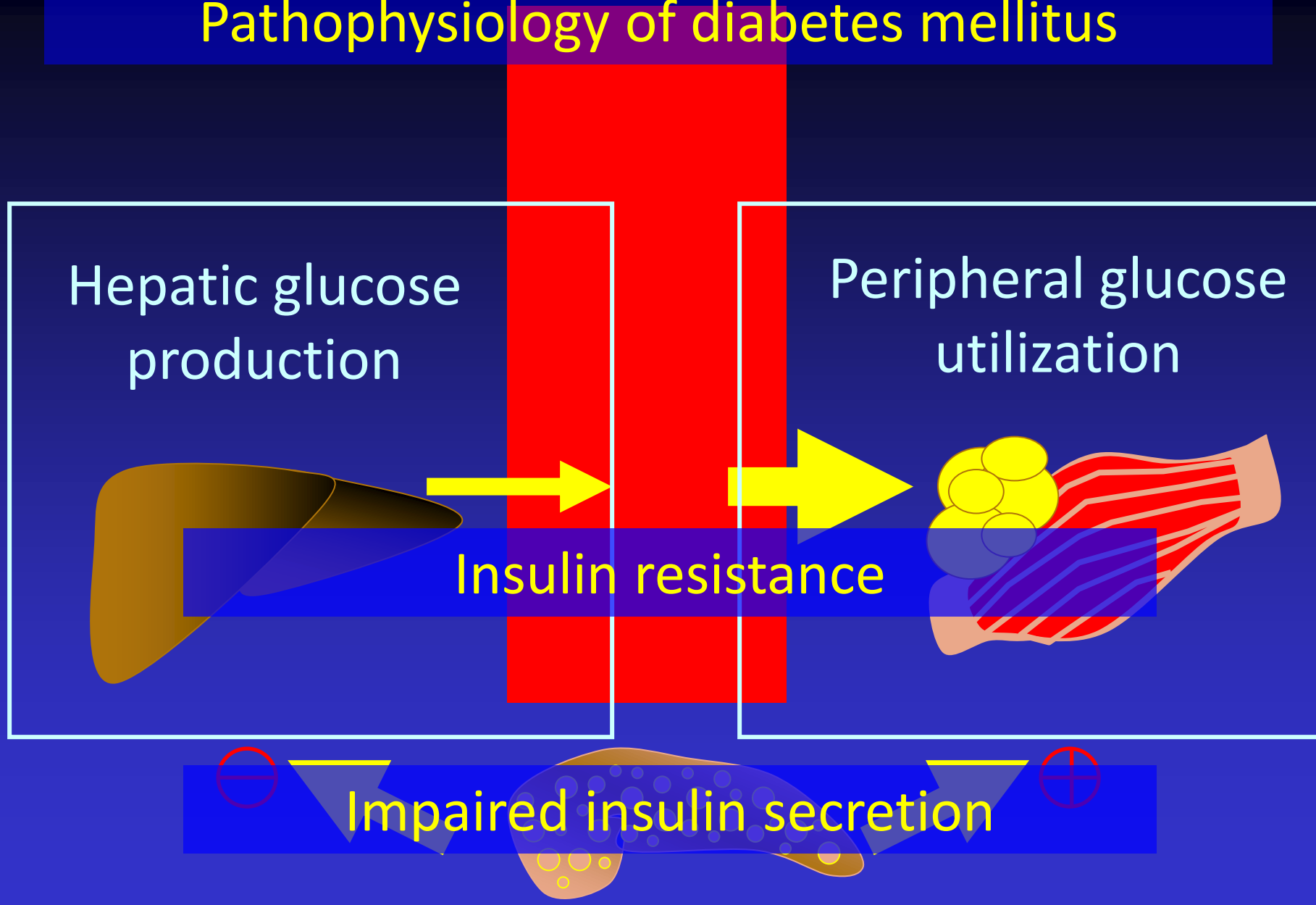
Glucose Homeostasis

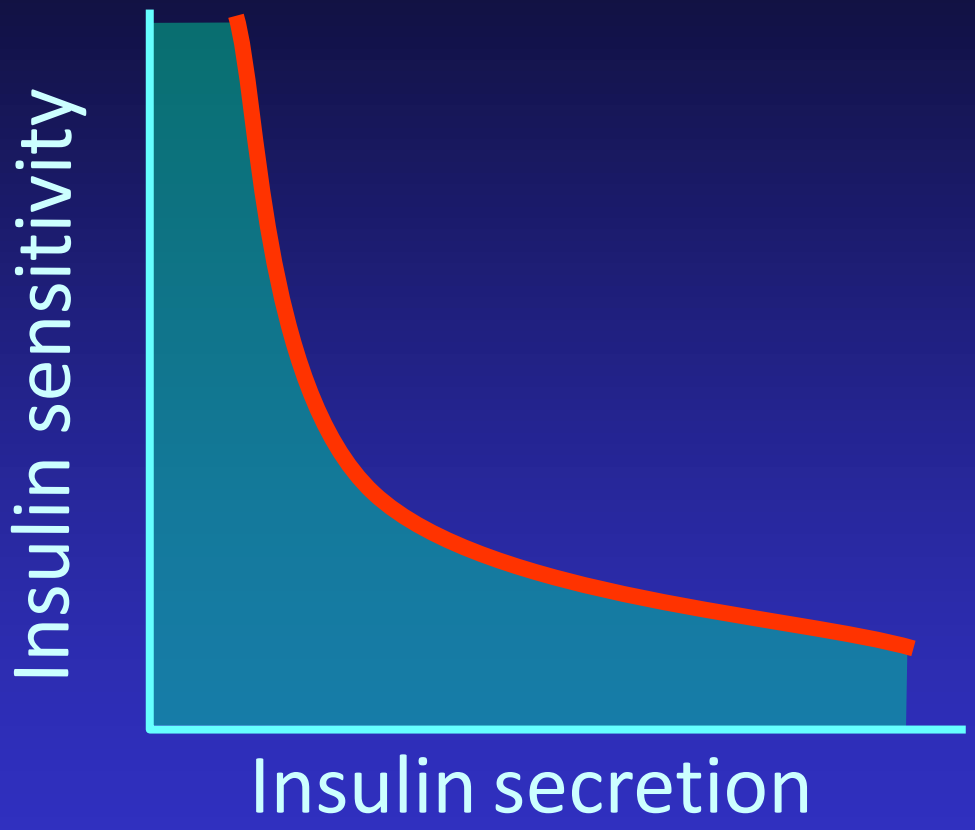

Oral intake
Hepatic glucose production
Peripheral glucose utilization
Glucose Homeostasis


Fasted stage
Hepatic glucose production
Peripheral glucose utilization


Fed stage
Oral intake
Hepatic glucose production
Peripheral glucose utilization

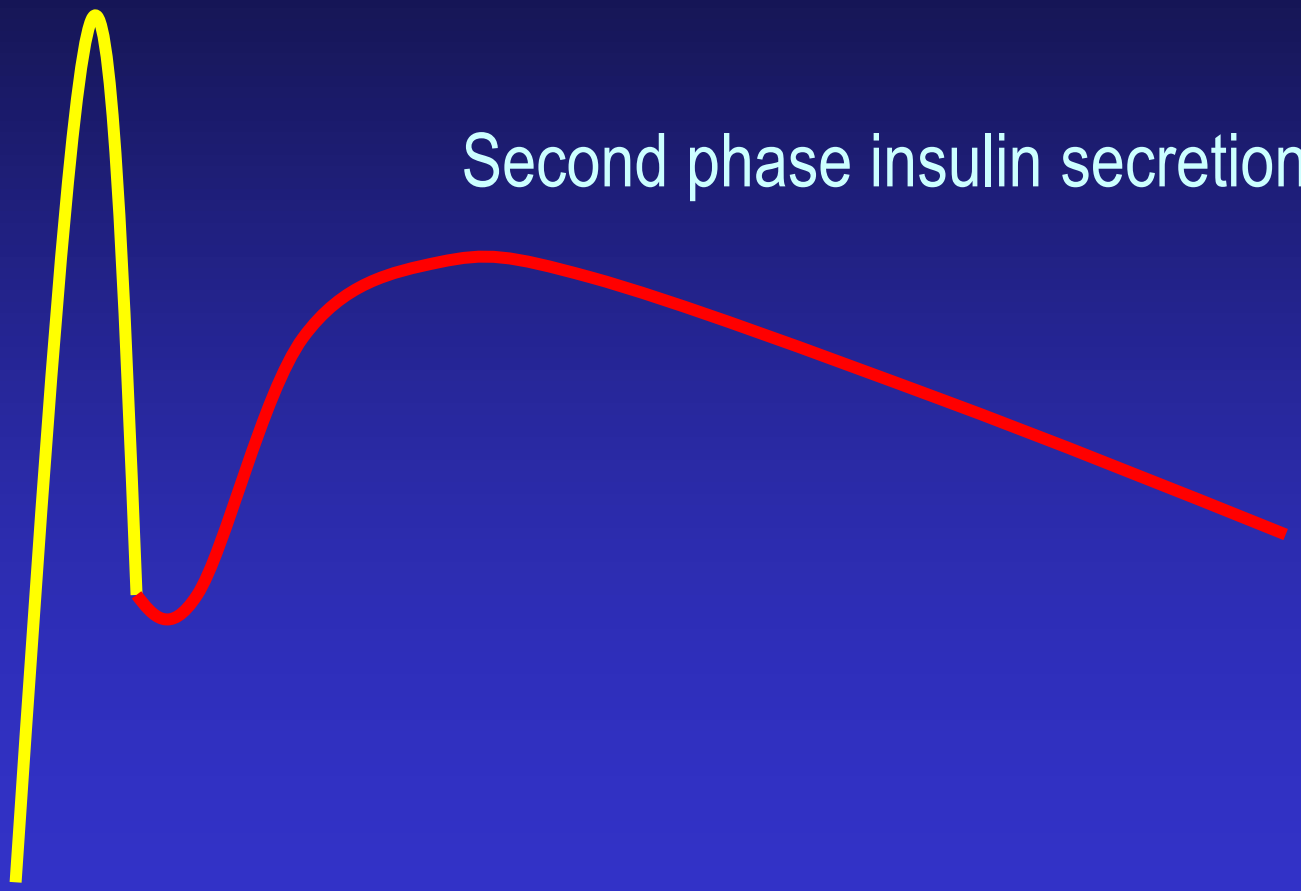
Pathophysiology of diabetes mellitus




Insulin sensitivity
Insulin secretion

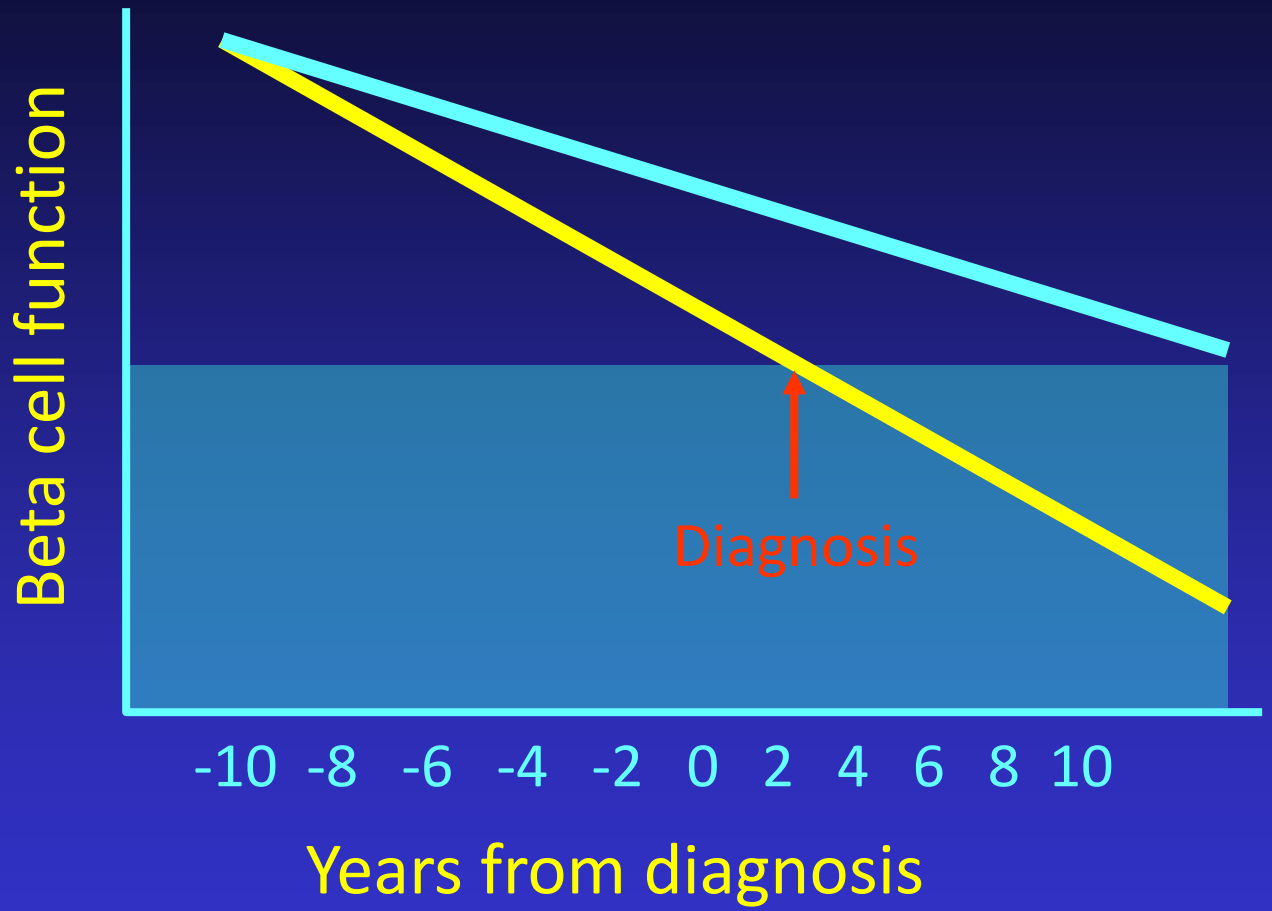
Insulin secretion

First phase insulin secretion



Progression of diabetes

	Insulin resistance	Insulin secretion		Plasma glucose
		First	Second	
1. NGT	↑	↑	↑	Normal
2. IGT	↑	↓	↑	Postprandial hyperglycemia mild fasting hyperglycemia
3. DM	↑↑	↓	↓	severe fasting hyperglycemia


Beta cell function
Diagnosis
-10 -8 -6 -4 -2 0 2 4 6 8 10
Years from diagnosis

Testing for diabetes in asymptomatic adults

1. Age $\geq$ 45

2. High risk groups

-First-degree relative with diabetes
-obesity (BMI >25kg/m^2)
-Hypertension
-HDL <35 mg/dl or TG >250mg/dl
-GDM/delivered a baby >4 kg
-PCOS
-Vascular diseases

Clinical manifestation

Symptoms:

polyurea, polydipsia, weight lost

Acute complications:

hyperosmolar, ketoacidosis, infections

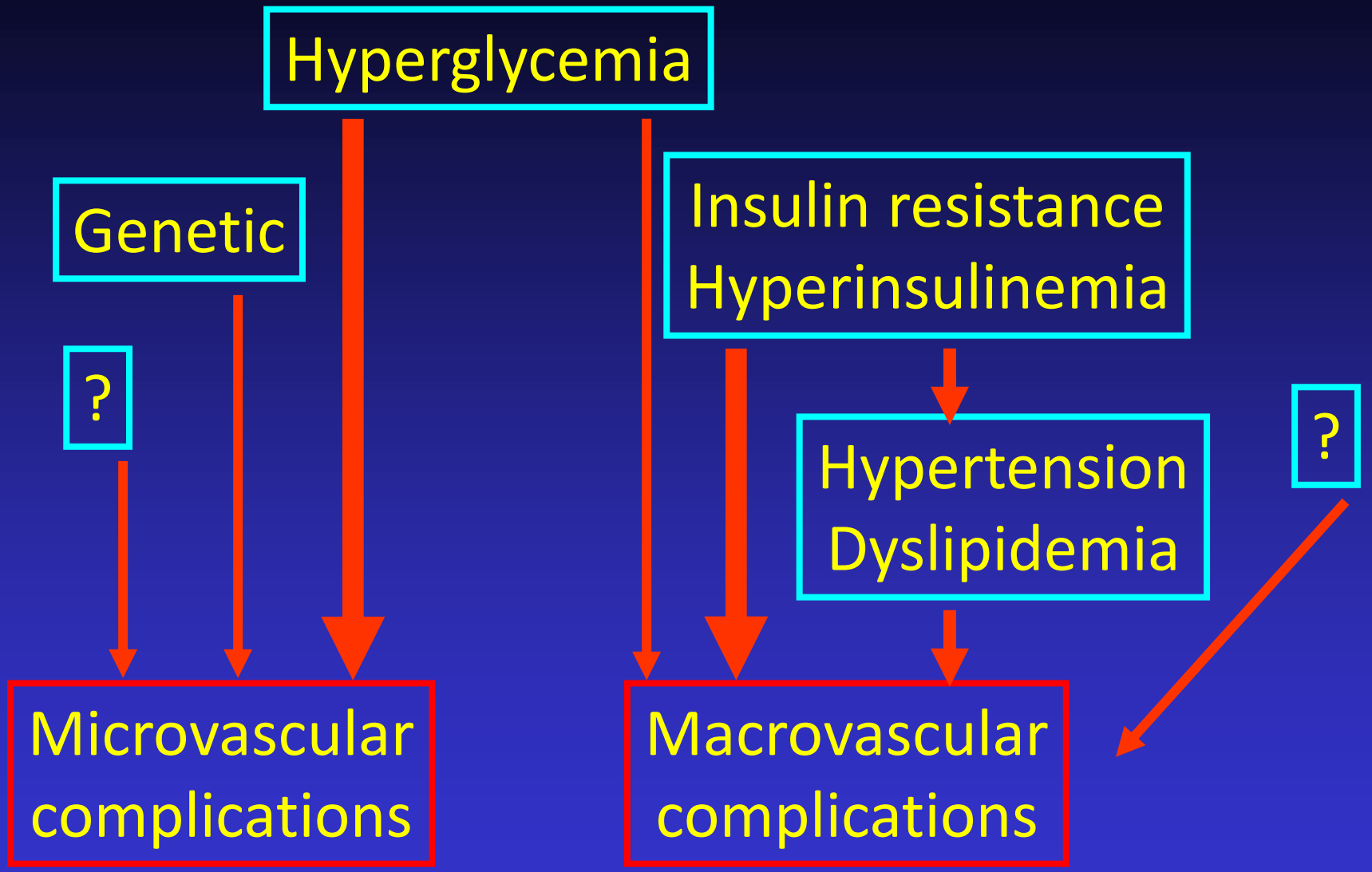
Chronic complications:

Microvascular:

nephropathy, retinopathy, neuropathy

Macrovascular:

cardiovascular, cerebrovascular, peripheral vascular diseases


Hyperglycemia
Genetic
?
Insulin resistance
Hyperinsulinemia
Hypertension
Dyslipidemia
?
Microvascular
complications
Macrovascular
complications

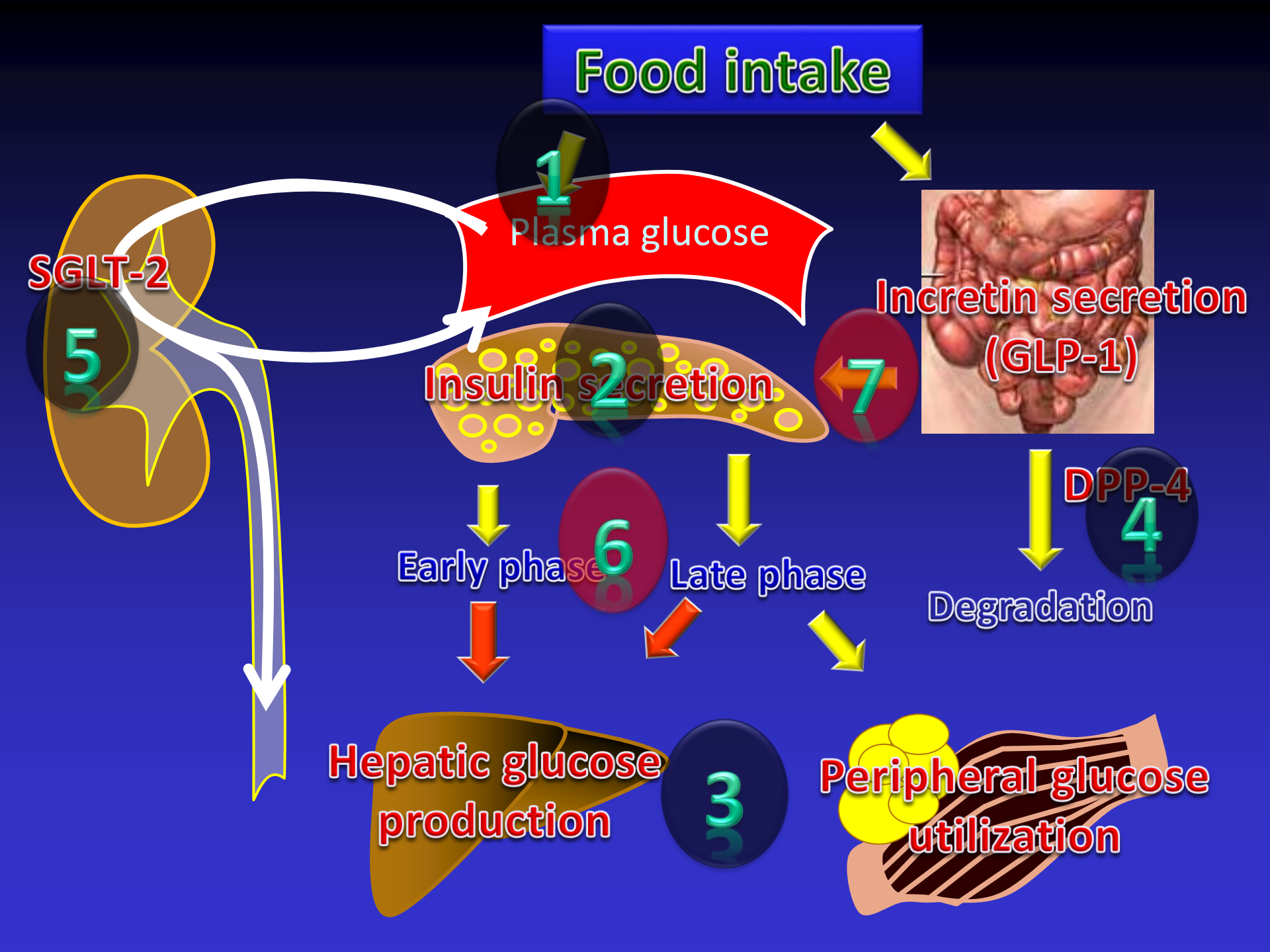
Initial evaluation

1. History:
diet, exercise, alcohol, smoking, drugs,
underlying diseases,

2. Physical examination:
BP, cardiovascular system, peripheral pulse, foot,
fundoscopic examination, oral examination, skin,

3. Lab:
plasma glcuose±HbA1c, Cr, lipid study,
UA, microalbumin, chest film, ±ECG

Diabetes Management


Food intake
1
Plasma glucose
SGLT-2
5
Incretin secretion
(GLP-1)
Insulin secretion
2
7
DPP-4
4
Degradation
Early phase
6
Late phase
Hepatic glucose
production
3
Peripheral glucose
utilization

Medications

Oral hypoglycemic agents

Insulin/GLP-1 receptor agonists

Diabetic management

1.GLYCEMIC CONTROL

2.PREVENT/TREAT CHRONIC COMPLICATIONS

3.TREAT CO-MORBID DISEASES

GLYCEMIC CONTROL

1. DIETARY CONTROL

2. WEIGHT REDUCTION/EXERCISE

3. MEDICATIONS

Medications

Oral hypoglycemic agents

Insulin/GLP-1 receptor agonists

Oral hypoglycemic agents

1. INCREASED INSULIN SECRETION

2. DECREASED INSULIN RESISTANCE

3. DELAYED INTESTINAL GLUCOSE ABSORPTION

4. INHIBIT URINARY GLUCOSE REABSORPTION

Increased insulin secretion

1. Sulfonylureas:

glibenclamide, glipizide,
gliclazide, glimeperide

Side effects:

hypoglycemia, weight gain

Increased insulin secretion

2. Glitinides:

repaglinide, nateglinide

Side effects:

hypoglycemia, weight gain

Increased insulin secretion

3. DPP-4 inhibitors:

sitagliptin, vildagliptin, saxagliptin, linagliptin, alogliptin

Decreased insulin resistance

1. Biguanide:

Metformin

Side effects:

GI side effects, lactic acidosis

Decreased insulin resistance

2. Thiazolidenedione:
pioglitazone

Side effects:

Fluid retention, weight gain
fractures, bladder cancer

Delayed intestinal glucose absorption

α-glucosidase inhibitors:
acarbose, voglibose

Side effects:
GI side effects

Inhibit urinary glucose reabsorption

SGLT-2 inhibitors:

canagliflozin
dapagliflozin
empagliflozin

Side effects:

Genitourinary tract infection,
Volume depletion, DKA

Indication for insulin

1. Type 1 diabetes mellitus
2. Gestational diabetes mellitus
3. Type 2 diabetes mellitus: if
    - 3.1. Pregnant
    - 3.2. Oral hypoglycemic agents failure
    - 3.3. Overt renal/hepatic failure
    - 3.4. Acute complications: DKA, HHS
    - 3.5. Acutely ill patients/ NPO

Insulin

Short acting insulin (RI):

actrapid, humulin R

Rapid acting insulin:

lispro, aspart

Intermediate acting insulin (NPH):

insulatard, humulin N

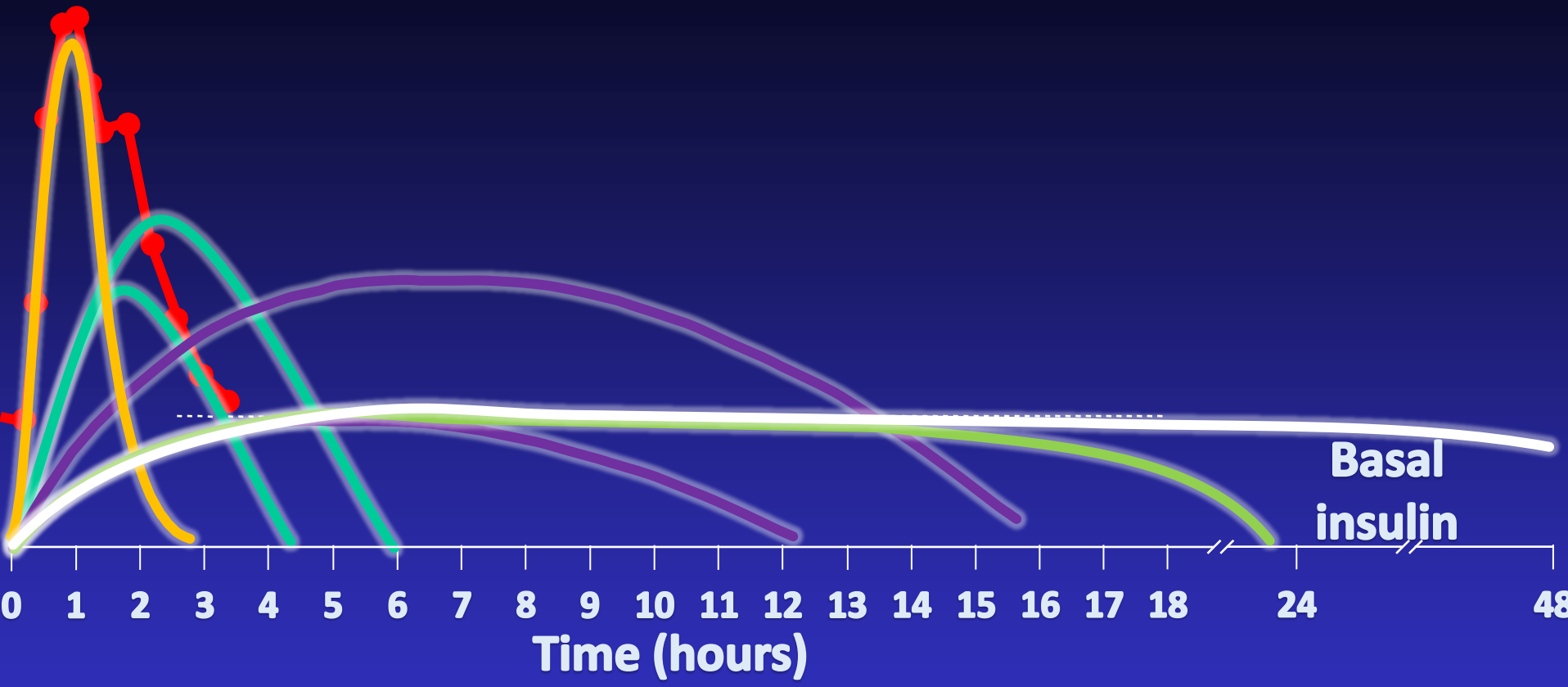
Long acting insulin:

glargine, detemir

Premixed insulin:

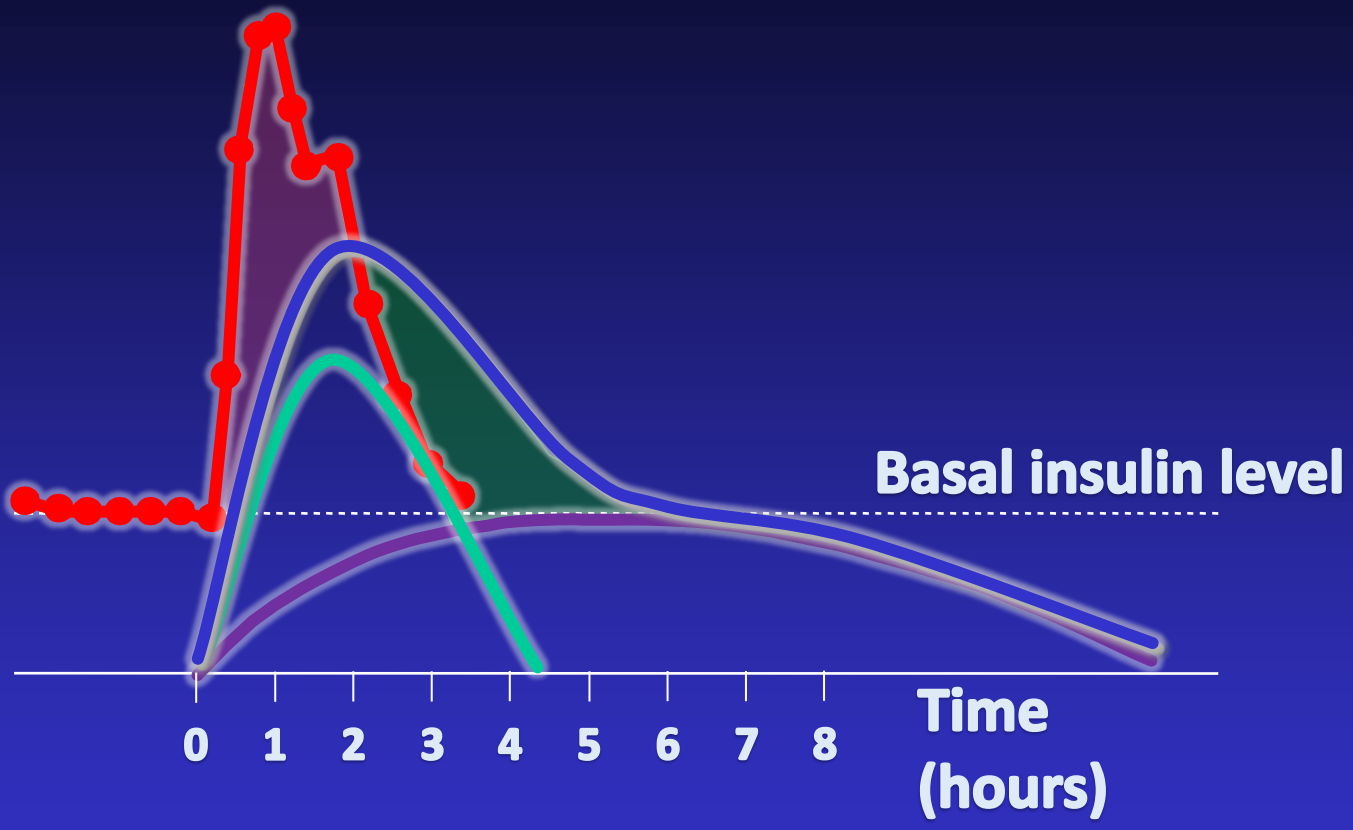
mixtard (30), humulin 70/30

Activity profiles of different types of insulin



- Physiological insulin profile
- Short acting insulin
- Intermediate acting insulin
- Rapid acting insulin
- Long acting insulin
- Ultra-long acting insulin

Insulin concept of premixed human insulin



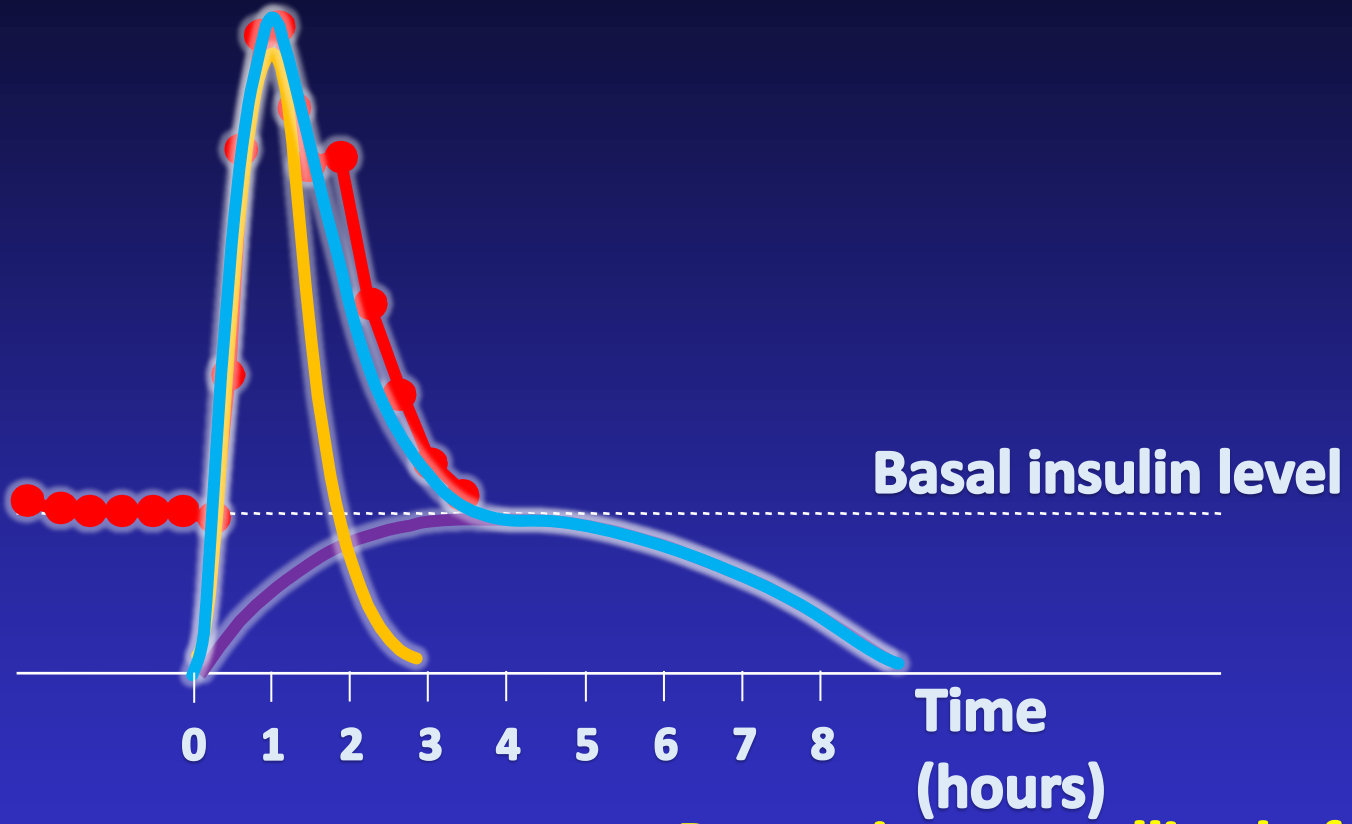
Physiological insulin profile

Intermediate acting insulin

Short acting insulin

Premixed human insulin

Insulin concept of premixed insulin analogue



Physiological insulin profile

Protamine-crystallised of rapid acting insulin

Rapid acting insulin

Premixed insulin analogue

GLP-1 receptor agonists

exenatide, liraglutide

Side effects:

GI side effects